Part A Chapter 6

From Interacting Individuals to Collective Patterns: The Emergence of Epidemics from Contact Behaviour[1]

1 Introduction

Microbes that sometimes enter the human body, such as viruses, bacteria, fungi and parasites, may have disturbing effects. Not seldom humans suffer from such infections and in the mean time propagate them to each other. Examples of types of infectious diseases are influenza, HIV, hepatitis, chlamydia, tuberculosis, and many others. The battle against such infections takes place both at the biological level in the body, and at the behavioural and social level. At the biological level, the human body has a sophisticated defence system against such disturbing influences from the environment; for example, see (Widmaier, Raff, and Strang, 2006, Ch 18, pp. 701-743). At the behavioural and social level, humans sometimes try to adapt their interaction behaviour to prevent propagation of infections from one human to another one. This chapter does not focus on the biological defence system, but on the propagation of infections in populations, in relation to the interaction behaviour, in particular the frequency and intensity of contacts that individuals have in the population. This is one of the many examples of a phenomenon where on the basis of individual decisions concerning one's own behaviour, collective patterns may emerge at the level of the population as a whole.

1.1 Domain description and focus

In this chapter the focus is on a population of individuals that can be in different states: susceptible (not infected yet), infective, or recovered (immune and not infectious). When an

[1] Based on: Jaffry, S.W., and Treur, J., Agent-Based and Population-Based Simulation: A Comparative Case Study for Epidemics. In: L.S. Louca, Y. Chrysanthou, Z. Oplatkova, K. Al-Begain (eds.), *Proc. of the 22th European Conference on Modelling and Simulation, ECMS'08*. European Council on Modeling and Simulation, 2008, pp. 123-130.

individual who is infective, has contact with another, susceptible individual then there is a chance that the other individual will also be infected due to this contact. This chance depends on the intensity of the contact. The overall chance that a susceptible individual is infected, also depends on the number of contacts with infective individuals. An example of a possible pattern, for example, for an easy transmittable infection such as influenza, is that the propagation goes so fast that in a few weeks time almost the whole population is infected. For example, if on the average one infective individual infects three others every day, then every day the number of infective individuals becomes multiplied by 4. An exponential growth pattern results, which leads, for instance, from one infective individual to 4^5 (which is about 1000) infective individuals in 5 days, and about one million in 10 days. In such a case the term epidemic is used to indicate the spreading of the infection over the population. Of course, such an unlimited exponential growth cannot continue, as after some time the bounds of the population will be reached, and the growth pattern will in fact be a pattern of limited growth, but then practically everybody may have become infected. For other types of infections, for example HIV or chlamydia, more intensive contacts (which usually occur less frequently) are needed for transmission, and therefore propagation may proceed slower.

An important question, especially for the more harmful infections, in a society is whether by measures at the behavioural and social level, it is possible to keep the number of infected persons in a population limited. And if so, how far should such measures go? It is clear that by avoiding any contact between individuals, propagation can be stopped, but that is not a realistic option. On the other hand, if there are still some contacts between individuals, will at the end the infection not be spread (perhaps by very slow propagation) over the whole population? Such questions are addressed in this chapter. To this end two types of models will be developed, one at the *individual behavioural level* (e.g., involving specific individuals and specific types of contact behaviours), and one at the *collective societal level* (e.g., involving the numbers of susceptible, infective and recovered persons and average types of contacts). Results of the two types of models will be compared.

1.2 Characteristic patterns to be explored

The following are patterns to be explored by the models:

- Does the initial phase of an infection show exponential growth of the number of infected persons?
- When infections transmit very easily, will the whole population become infected according to a logistic growth pattern of the number of infected persons?
- In how far is it possible to adapt contact behaviour and the number and frequency of contacts to a level low enough to avoid propagation of the infection over the whole population? Is there a nonzero threshold for the amount of interaction such that under this threshold the infection dies out?

2 Analysis of the Main Aspects and Their Relations

This section describes globally the relevant concepts used and their relations. The analysis of epidemics has a long history, going back, for example, to (Ross, 1916; Kermack and McKendrick, 1927). More recent presentations can be found in (Anderson and May, 1992) and (Ellner and Guckenheimer, 2006, Ch. 6, pp. 183-215).

2.1 Identification of relevant concepts

First of all, a distinction has to be made between *non-infected* and *infected* individuals (*susceptibles* vs. *infectives*), the latter of which are infectious for the former. A third class of individuals may consist of those that already were infected, but have *recovered* and therefore are immune and not infectious anymore, based on a *recovery rate* (in the literatrure sometimes also persons that have died are included here). Furthermore the *frequency* of *contacts* (per day, the time unit chosen) plays a main role, which can be related to being at the same *location*; the chance that in a contact *infection (transmission)* occurs depends on the *contact intensity*. Note that these concepts can be used both at the individual behavioural level (e.g., specific infected individuals and types of contact behaviours), and at the collective societal level (e.g., statistics of the numbers of infected persons and average types of contacts). In summary, the following concepts are used in the model:

- susceptibles: non-infected individuals
- infective individuals
- recovered individuals: immune and non-infective (also dead individuals can be considered here)
- infection transmissions (per day)
- contacts (per day)
- recovery rate
- contact intensity
- contact frequency
- location

2.2 Overall structure of the model

The model is described at two levels:

- the *collective societal level* of the population as a whole, and the sizes of subpopulations of susceptible, infective and recovered persons
- the *individual behavioural level* of the various behaviours of specific persons and their interactions.

At the collective level the functioning of the society is described, for example, with respect to statistics for numbers of infected persons, and consequences for health care organisation. At the level the individual behaviours are described: behaviours of individuals and contacts between individuals due to these behaviours.

Interaction between the individual and collective level takes place in two directions:

- *Upward interaction*
  The behaviours by the various individuals affect the society's functioning, including, for example, processes for health care organisation
- *Downward interaction*
  The society's states and processes over time affect the individual behaviours and interactions, for example, information about the occurrence of an epidemic, or measures such as closing schools to decrease the number of contacts, vaccination programs, or globally broadcasted advices for behaviour adaptations.

This model structure shows how phenomena at the collective societal level can interact with processes of individuals at the individual behavioural level.

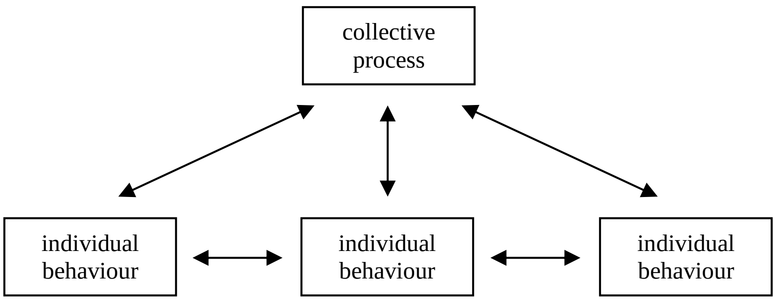

Figure 1 Overall structure of the model

2.3 Relationships between the identified concepts

The temporal relationships between the identified concepts are as follows.

- infectives, susceptibles, and contact frequency affect the relevant contacts (per day)
- relevant contacts (per day) and contact intensity affect the infection transmissions (per day)
- infection transmissions per day affect infectives and susceptibles
- recovery rate affects infectives and recovered individuals

Note that these relationships can be described for the individual level (for individuals and their particular infection states) and for the level of the society as a whole (for numbers of susceptibles, infectives and recovered persons). In graphical form these temporal relationships are depicted in Figure 2.

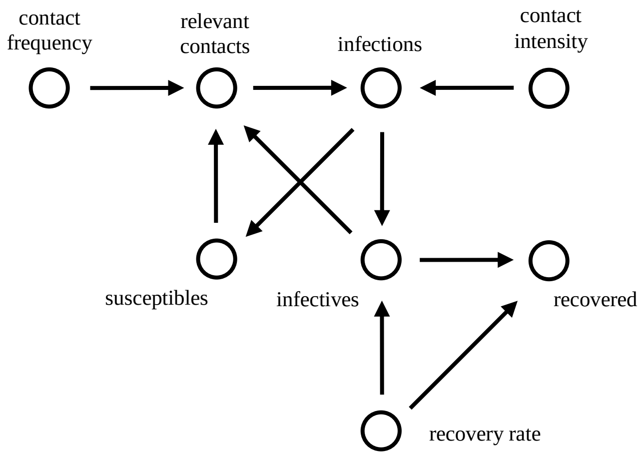

Figure 2 Dynamic relationships in the model

3 The Detailed Model

In this section the detailed model is described, by formalising the relevant concepts and their dynamic relations in a detailed manner for both the collective and individual level.

3.1 Formalising the concepts used

To obtain a detailed model, first the concepts used are formalised. This is done both at the level of the specific individual behaviours and at the collective societal level of the population as a whole and the subpopulations of noninfected, infected and immune persons.

Formalisation of concepts at the collective level

At the collective level, at each point in time the sizes of the three populations are formalised by variables over the real numbers. Similarly, the numbers of relevant contacts and new infections taking place per day (the time unit chosen) are formalised by variables over real numbers. Three constant parameters (also real numbers) are used for contact frequency, contact intensity and recovery rate. For an overview, see Table 1.

number of susceptible individuals (non-infected)	*Susceptibles*
number of infective individuals	*Infectives*
number of recovered individuals (immune or dead, non-infective)	*Recovered*
number of relevant contacts per day (infective with susceptible)	*RelevantContacts*
number of infections taking place per day	*Infections*
contact frequency per person per day	*ContactFrequency*

contact intensity	*ContactIntensity*
recovery rate (fraction of infectives recovering per day)	*RecoveryRate*

Table 1 Formalisation of the relevant concepts: collective level

No concept for location is used, as contacts are directly based on the sizes of the populations and the contact frequency.

Formalisation of concepts at the individual level

At the individual level, for the relevant concepts a numerical variable is introduced that takes real or integer numbers as values. At every time point each individual is at some location, at random. Contacts between individuals are modelled as being at the same location. By taking the number L of locations (numbered by $1, 2, ..., L$) lower or higher, a specific contact frequency is modelled. The three infection states are also numbered by 0 (susceptible), 1 (infective), and 2 (recovered). Notice that the last three rows, where total numbers for subpopulations are represented, are in fact not part of the model at the individual level, but they are aggregated concepts that have been added to allow comparison with the model at the collective societal level. They are determined by just counting the individuals in a specific infection state (susceptible, infective, recovered, respectively). For an overview, see Table 2.

susceptible infective recovered	*InfectionStateA1 = 0, InfectionStateA2 = 0, ……* *InfectionStateA1 = 1 , …* *InfectionStateA1 = 2 , …*
location of individuals	*LocationA1 = 1, LocationA1 = 2, ……* *LocationA2 = 1, LocationA2 = 2, ……* ….
contact frequency per day for one individual	*ContactFrequency*
contact intensity	*ContactIntensity*
recovery rate (chance per day for infective to recover)	*RecoveryRate*
number of susceptible individuals	*Susceptibles*
number of infective individuals	*Infectives*
number of recovered individuals	*Recovered*

Table 2 Formalisation of the relevant concepts: individual level

3.2 Formalising the relationships

The dynamics of these concepts involve temporal relationships, which are analysed in more detail below. They describe the basic parts of the process; see also Figure 2. First the relationships at the collective societal level are addressed.

Dynamic relationships formalised for the collective level

Susceptible, infective and recovered populations, contacts and contact intensity affect the number of infections taking place

Each susceptible person has (per day) a number of contacts indicated by *ContactFrequency*. From these contacts a fraction *Infectives(t)/N* is with infective individuals, where *N* is the (fixed) number of the three populations together. Therefore the number of relevant contacts per day is:

$$RelevantContacts(t) \quad = \quad ContactFrequency*Susceptibles(t)*Infectives(t)\ /N$$

Moreover, in a fraction of the contacts the infection is transmitted. This fraction is what is meant by *ContactIntensity*; therefore the number of new infections per day is:

$$Infections(t) \quad = \quad ContactIntensity*RelevantContacts(t)$$

Infections and recovery rate affect non-infected, infectious, and immune populations

Given the number of infections *Infections(t)* per day, in the time interval between t and $t+\Delta t$ the number of infections is $Infections(t)*\Delta t$. This is subtracted from the susceptible population, and added to the infective population. Furthermore, the number *RecoveryRate* indicates the fraction of the infective population per day that becomes recovered (and not infective anymore). Therefore over the interval between t and $t+\Delta t$ a number of $RecoveryRate*Infectives(t)*\Delta t$ is taken from the infective population and added to the recovered population. Therefore the following temporal relationships are used.

$$Susceptibles(t+\Delta t) \quad = \quad Susceptibles(t) \ - \ Infections(t)*\Delta t$$

$$Infectives(t+\Delta t) \quad = \quad Infectives(t) \ + \ (Infections(t) \ - \ RecoveryRate*Infectives(t))*\Delta t$$

$$Recovered(t+\Delta t) \quad = \quad Recovered(t) \ + RecoveryRate*Infectives(t)*\Delta t$$

Note that by these relationships the sum of the three populations always remains the same *N*: what adds to the recovered population subtracts from the infective population, and what subtracts from the susceptible population adds to the infective population.

In Table 3 based on these relationships, for an example scenario the dependencies between the values are shown.

t	susceptible	infective	recovered	relevant contacts	new infections	contact frequency	contact intensity	recovery rate
0	99	1	0	0.8	0.4	0.8	0.5	0.1
1	98.6	1.3	0.1	1.0	0.5	0.8	0.5	0.1
2	98.1	1.7	0.2	1.3	0.7	0.8	0.5	0.1
3	97.4	2.2	0.4	1.7	0.8	0.8	0.5	0.1
4	96.6	2.8	0.6	2.2	1.1	0.8	0.5	0.1

5	95.5	3.6	0.9	2.7	1.4	0.8	0.5	0.1

Table 3 Dependencies between the different concepts over time: collective level

Dynamic relationships formalised for the individual level

From contact frequency to locations

Contact between two individuals takes place by being at the same location. Therefore the number of locations has a direct relationship with the contact frequency. If *L* is the number of locations, and *N* the number of individuals, then the average number of individuals on one location is *N/L*, so the average number of contacts of one individual at such a location is *N/L – 1*. This has to be equal to the contact frequency. Therefore

ContactFrequency = N/L - 1

gives the relation between number of locations and contact frrquency. To be able to compare this model at the individual level to the model at the collective level, it is convenient to have contact frequency as a basic parameter. To this end, the relation between the number of locations and contact frequency shown above is used in an inverse manner to determine the number of locations for a given value of the contact frequency:

L = N/ (ContactFrequency + 1)

For a given contact frequency, this *L* is taken as a bound for the number of locations: the locations are indicated by the natural numbers *l* with $1 \leq l \leq L$. Locations of the individuals are determined at random, using this bound *L*, as follows:

LocationA1(t) = RandBetween(1, N/ (ContactFrequency + 1))

Based on this at every point in time for each individual a new location from *1* to *L* is determined by taking at random one of the natural numbers between *1* and *L*: *RandBetween(1, L)* just gives a random natural number between *1* and *L*.

From infection states, locations and contact intensity to changed infection states

The following relationships describe the changes of the infection state of individual A1. When a susceptible person is at a location where one or more infective persons are present, the transmission of the infection at that time point has a probability given by the contact intensity. Moreover, for someone who is infective there is a probability of recovery given by the recovery rate. This is modelled by:

InfectionStateA1(t+Δt) = 1

if

InfectionStateA1(t) = 0 and *r1 < ContactIntensity* and
for some *k* it holds *LocationA1 = LocationAk* and *InfectionStateAk = 1*
or *InfectionStateA1(t) = 1* and *r2 ≥ RecoveryRate*

Here *r1* and *r2* are two independent random numbers between *0* and *1: RandBetween(0, 100)/100,* fixed per time point, but refreshed at new time points.

Moreover,

InfectionStateA1(t+Δt) = 2

if

InfectionStateA1(t) = 1 and *r2 < RecoveryRate*
or *InfectionStateA1(t) = 2*

In all other cases

InfectionStateA1(t+Δt) = 0

Similarly the infection state of the other individuals changes.

From infection states of individuals to numbers of individuals with a certain infection state
The numbers of individuals with a certain infection state are just counted.

Susceptibles(t) = number of individuals *Ak* for which *InfectionStateAk(t)* = *0*
Infectives(t) = number of individuals *Ak* for which *InfectionStateAk(t)* = *1*
Recovered(t) = number of individuals *Ak* for which *InfectionStateAk(t)* = *2*

Such counting can be done using summation over conditional values; for example,

Susceptibles(t) = Σ_k *if(InfectionStateAk(t) = 0, 1, 0)*

Where the expression *if(condition, value1, value2)* gives *value1* if *condition* is true, and *value2* else.

4 Simulation Experiments

A number of simulation experiments have been performed using Microsoft Excel, both at the collective societal level and the level of the individual behaviours.

Simulation examples at the collective societal level

In Figure 3 results are shown of one of them, with time scale in days. In the first simulation shown in Figure 3 the whole population gets infected; it used the following parameter settings:

ContactFrequency 0.8
ContactIntensity 0.5
RecoveryRate 0.05

Note that in this and next figures the scales at the vertical axis differ. The number of noninfected persons decreases to zero, while the number of infectious persons increases until day 20 and decreases after this day. The number of immune persons shows a logistic growth pattern with the whole population of 100 as limit.

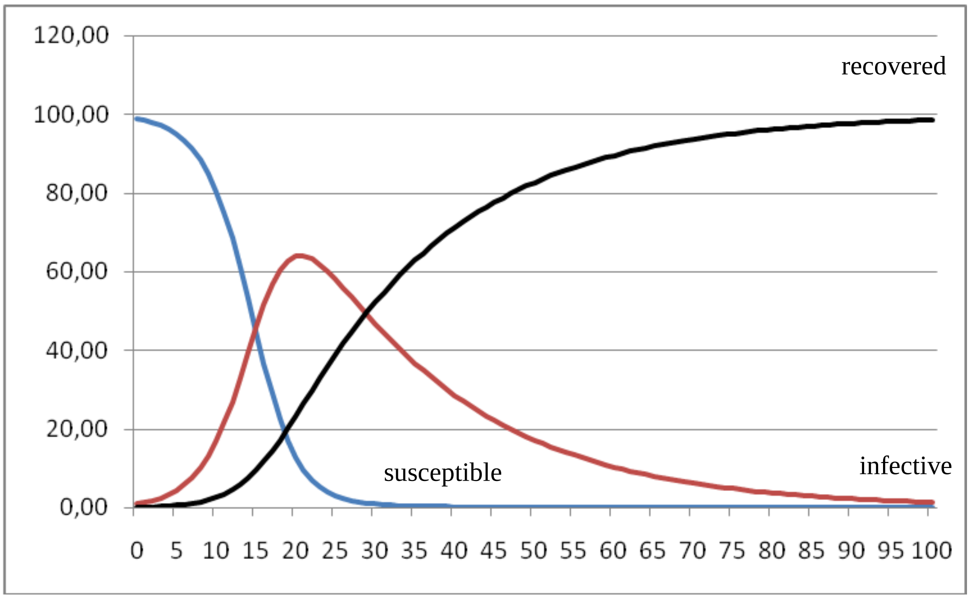

Figure 3 Pattern in which the whole population gets infected: collective societal level

In the second simulation, shown in Figure 4, only part of the population gets infected. The parameter settings were:

ContactFrequency	*0.6*
ContactIntensity	*0.2*
RecoveryRate	*0.1*

Apparently here the contact frequency and intensity were low enough to let the infection die out: more than 65% of the population is never infected (upper graph in Figure 4). The logistic growth pattern of the (infected and) recovered population has its limit around 35 (lowest graph in Figure 4). Nevertheless, the individuals still did not bring their contacts down to zero, or even close to zero. This shows that by relatively small differences in behaviour at the individual level, relatively big differences at the collective level can be realised.

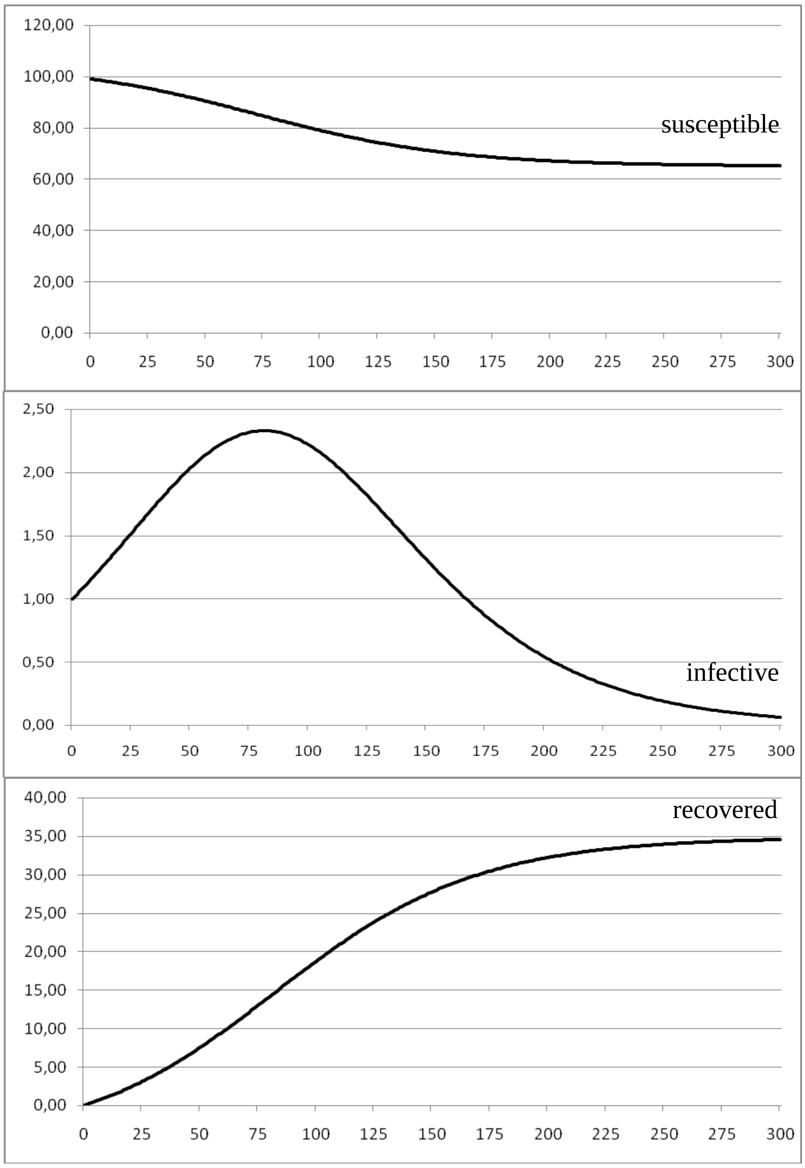

Figure 4 Pattern in which less than 35% of the population gets infected: collective societal level

Simulation examples at the level of the individual behaviours

Similar simulation experiments as the ones described above have been performed using the model at the level of the individuals. As this model is based on random choices, the patterns can vary.

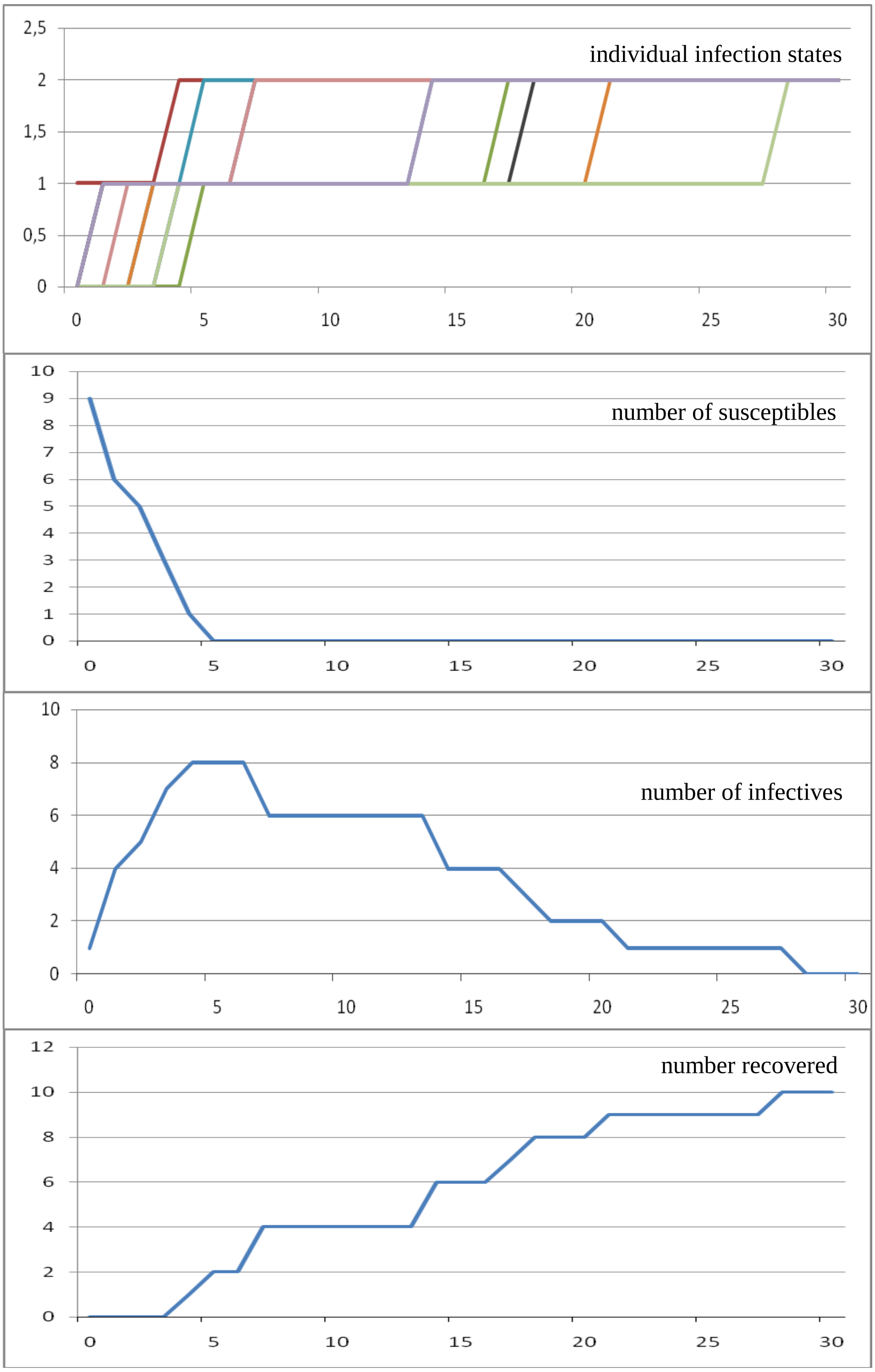

Figure 5 Pattern in which the whole population gets infected: individual level

In Figure 5 an example trace based on the following parameter settings is shown.

ContactFrequency	*0.8*
ContactIntensity	*0.5*
RecoveryRate	*0.05*

These settings are the same as for the trace shown in Figure 3. In the upper graph it is shown how individuals get infected one after the other. At the start A2 is infective. Already from the second day on, A4, A7 and A10 get infected. After that A8, A5 and A6 follow, and finally A1, A9 and A3 get infected. In the second graph in Figure 5 the number of susceptibles is shown, in the third graph the number of infectives, and in the lower graph the number of recovered individuals. The pattern is similar to the pattern shown in Figure 3. Also in other simulations by the individual level model this pattern occurs.

In Figure 6 an example trace based on the following parameter settings (the same as for the trace in Figure 4) is shown:

ContactFrequency	*0.6*
ContactIntensity	*0.2*
RecoveryRate	*0.1*

In the upper graph it is shown how three individuals get infected. At the start A2 is infective. Soon A6 gets infected but recovers already in two days. Since A2 takes longer to recover, A8 is infected on day 5. After 6 days A8 recovers, and in the meantime also A2 recovered. No further infections took place. In the second graph in Figure 6 the number of susceptibles is shown, in the third graph the number of infectives, and in the lower graph the number of recovered individuals. The pattern is similar to the pattern shown in Figure 4. However, from other simulations it was found out that this example trace is a bit exceptional for this setting. Most traces of the individual model show either only one or two infectives, after which the epidemic dies out, or (almost) all individuals become infected. See Figure 7 for an overview of 100 experiments. The average number of recovered persons for this sample is 5.71. Note that this means that the model at the collective level shows a kind of average pattern that for the model at the individual level almost never occurs.

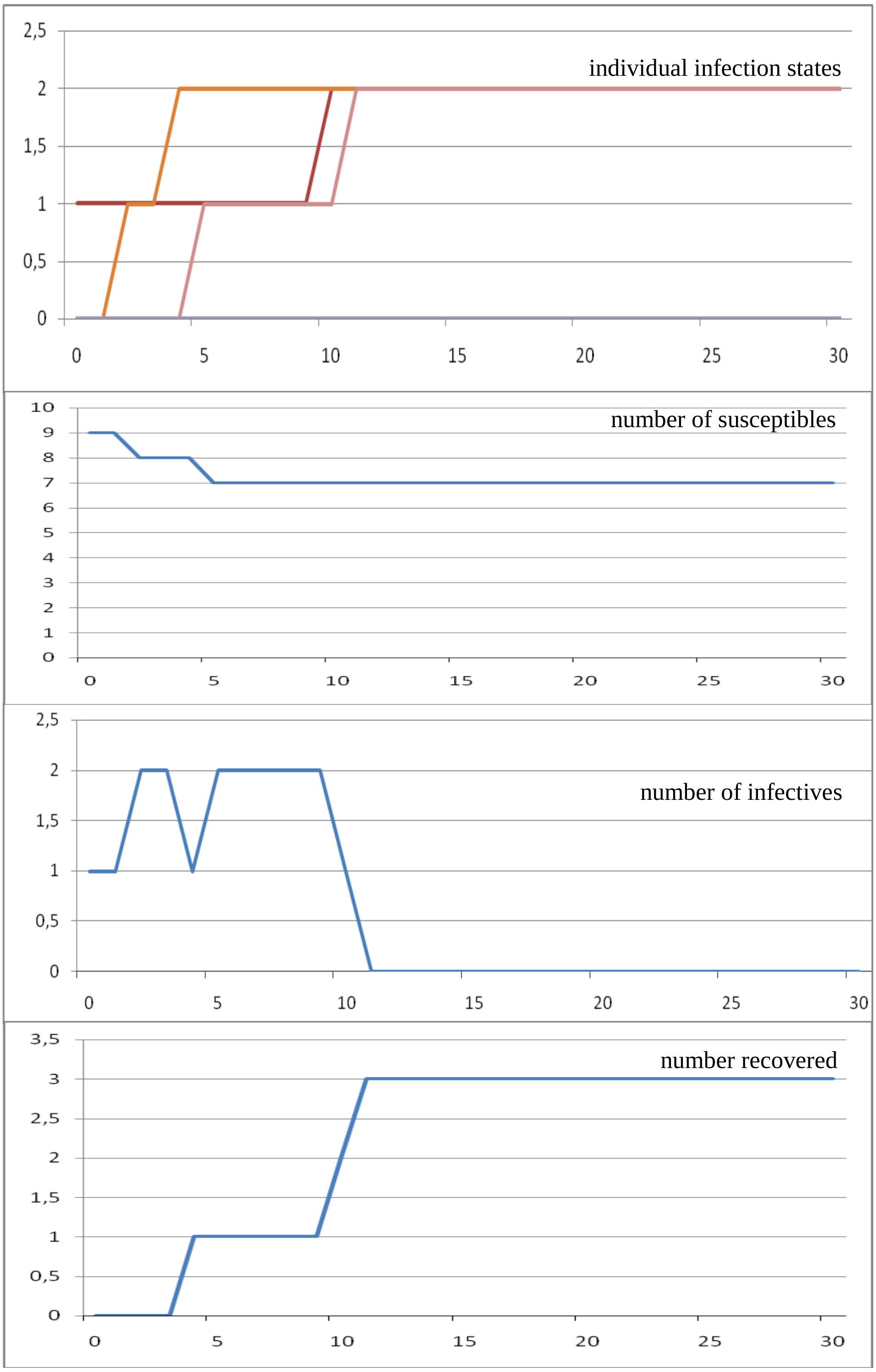

Figure 6 Pattern in which only part of the population gets infected: individual level

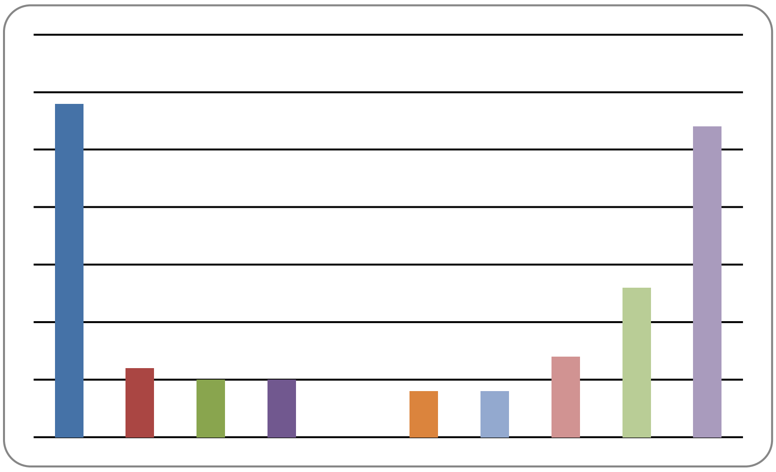

Figure 7 Overview of the number of recovered persons for a sample of 100 simulations

5 Evaluation

In this section, the simulations performed using the model are evaluated with respect to the characteristic patterns identified in Section 1.2. Moreover, the assumptions behind the model are summarised. Finally, possibilities of variations or refinements of the model are discussed.

5.1 Evaluation of the simulations and characteristic patterns

The following characteristic patterns were formulated in Section 1.2. For each of them it is discussed in how far they are realised by the model.

Does the initial phase of an infection show exponential growth of the number of infected persons?

The graphs for the infective and recovered persons in Figures 3 to 6 initially show exponential growth.

When infections transmit very easily, will the whole population become infected according to a logistic growth pattern of the number of infected persons?

In the cases described in Figures 3 and 5 indeed the whole population gets infected through a logistic growth pattern.

In how far is it possible to adapt contact behaviour and the number and frequency of contacts to a level low enough to avoid propagation of the infection over the whole population? Is

there a nonzero threshold for the amount of interaction such that under this threshold the infection dies out?
The graphs in Figures 4 and 6 show this possibility. Where exactly the threshold can be found in terms of values for the three parameters (contact frequency, contact intensity, recovery rate) is still to be explored.

5.2 Assumptions behind the models

The following assumptions were made.

- The population is fixed
- Every individual in principle can have contact with every other individual
- Contact frequencies and intensities are uniform over individuals (and static)
- In the model at the individual level, the probability to become infected is independent of the number of infective persons at the same location.
- No cognitive decision processes are taken into account for contact behaviour
- Individuals that are recovered cannot be infected again
- No vaccination programmes are organised.

5.3 Possible variations and refinements

Variations and refinements of the model into a more complex model can be obtained, for example, in the following manners; see also (Ellner, and Guckenheimer, 2006, Ch. 6, pp. 183-215).

- Make a large number of simulations by the model at the individual level and compare the results with the model at the collective level.
- Find theshold values where the epidemic dies out.
- In the model at the individual level, the probability to become infected at a certain time point can be made dependent on the number of infective persons at the same location. If this number is k, then the probability of not being infected by one of them is $(1 - ContactIntensity)$, so assuming independence, the probability of not being infected by any of them is $(1 - ContactIntensity)^k$; therefore the probability to get infected at that location at that time point is $1 - (1 - ContactIntensity)^k$.
- Incorporate individual differences in contact behaviour. As a specific case, consider a (core) population that has higher contact frequencies and/or intensities than other individuals.
- Incorporate a fourth population of isolated infective individuals that are no longer available for contacts.
- Consider the case that recovered individuals become susceptible again: no immunity, or immunity decays after some time.
- Consider the typical children's diseases, where all the time new births are adding individuals to the population.

- Explore what can be done by vaccination programs.
- Incorporate change of contact behaviour of individuals depending on the number of infectives. For example, if the number of infectives becomes higher, the contact frequency and/or contact intensity become lower; see also (Xiao and Ruan, 2007).
- Develop a model for transmission of an influenza infection from birds to humans; see also (Iwami, Takeuchi, and Liu, 2007).
- Incorporate internal cognitive models for decision making about contact behaviour; for example, based on beliefs, desires and intentions.

References

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